Project Suggestions

Evaluating Historical Text Corpora

revision of April 12, 2026

For M.EP.11b, this seminar entails a personal project in accordance with the project specification. The present document serves to indicate some possible avenues of research to that end.

Poetic Formulas

The course demo notebook on *n*-grams offers some very rough guidance to methods of quantifying collocations present in a text corpus; the Battles article listed in the syllabus bibliography constitutes a more thorough implementation applied to the corpus of Old English poetry, which you might want to test or improve using your own methods. CLASP is an online corpus that identifies formulas in Old English poetry, but without offering its pattern database for download. The identification and definition of poetic formulas is a worthwhile and feasible research question, in which perhaps the main challenge will be the distinction between sequences (near-)exclusive to the poetry and those occurring also in the prose. A project along these lines should therefore investigate a sizeable corpus containing both prose and verse, such as (for Old English) the YCOE–YCOEP pair of annotated corpora, or the former alongside the ASPR corpus, or DOEC. The same methods may not be as fruitful when applied to Old Norse inasmuch as the syllable count of skaldic poetry is so tightly regulated that it uses a more compressed language, resulting in less overlap of forms with the prose; but that is not to say it is not worth attempting (using documents from heimskringla.no and/or Menota, and using the Skaldic Project database for corpus orientation). The principle may likewise be applied to any choice of Middle English or Latin prosimetrical bicorpus, or any sizeable corpus of early Modern poetry.

Language Alignment: A Comparative Approach

Old Low German (i.e. Old Saxon) and Old English are very similar both linguistically and in terms of their poetic conventions and formulas. Whereas the linguistic connection is plausibly explained by the fact that the former served as perhaps the main parent language to the latter, we may legitimately ask how they came to have such similar poetic traditions after centuries of geographic separation, particularly in the light of English missionary activity on the Continent in the intervening time. We have fragments of a OLG poem (the *Old Saxon Genesis*) from which the OE *Genesis B* must have been copied or translated, but the longest OLG poem, the *Heliand*, is a richer source of poetic lexicon and formulas in view of its length of just under 6,000 lines, or nearly two *Beowulfs*. The text of one *Heliand* manuscript is available with lemmatization, part-of-speech (POS), and syntactical markup, as the result of the HeliPaD project (see the demo notebook on parsed corpora on how to load all but the syntactical structure of this corpus), and its lemmatization and POS tagging has recently been extended to the other manuscript witnesses (see the repository behind the Helix project). There is thus potential here for attempts to identify shared stems and formulas between the two poetic corpora. Using traditional NLP methods, this would likely have to involve a rule set for translating between the phonological divergence between the two languages, and probably two sets of stemming rules, combined with a Levenshtein comparison of similar forms; it could also involve an effort to map OLG lemmas onto OE forms. Transfer learning as used in large language models would likely be more effective, and you are free to loop in such methods if you can use a Python wrapper (either CLTK 2.0 or some more immediate LLM API) to do so, and if you can meaningfully reflect on the approach and its reliability.

Source Analysis: Language Alignment and Other Approaches

Another bicorpus approach is to involve a foreign-language corpus used as a source quarry; in the Western Middle Ages, that source corpus is usually Latin. Some online text corpora cover Latin originals alongside their translations on a sentence-by-sentence basis. For instance, DOEC encodes the Latin originals ("lemmas") alongside their Old English translations for its subcorpus of glosses, most notably the fourteen continuous Psalter glosses organized by Bible verse. ECHOE furthermore offers sentence-by-sentence sources of Old English homilies, but it does not make them available for download or include them in its repository. My own attempts to align the Latin and Old English Psalter glosses at the word level have suggested they are too limited a corpus to infer a Latin–Old English dictionary of any reliability, but more research is surely needed. An approach sometimes attempted is to transfer word embeddings between corpora in two different languages. But a straightforward TF-IDF angle on sources used may be more promising where small corpora are concerned, using known Latin sources to find additional sources without ever having to align the different languages. For instance, many of the Latin sources used in medieval texts were published as the series usually referred to as Patrologia Latina (PL), which is available as part of Corpus Corporum. A TF-IDF search engine comparing Latin sequences in e.g. Old English texts with all the documents contained in PL would be a valuable resource.

Reconstructing Textual Damage

One avenue of research that might not have a high likelihood of success, but offers a worthwhile challenge all the same and need not be overly difficult to develop depending on the approach chosen, is the probabilistic reconstruction of text lost due to manuscript damage. A corpus like ECHOE is sufficiently large to allow the recognition of frequent patterns (see Poetic Formulas above, and the demo notebook on *n*-gram analysis). Since it also encodes where text is lost and approximately how much, it should be possible to develop a system that takes as its input a window of context tokens and the approximate size of the gap in characters, and predicts what text might be missing judged by common formulas or next-token-prediction. This could even be achieved using a rule-based approach, simply listing passages similar to the surviving text and highlighting the material that could fill the gap. A worthwhile document on which to test this approach is ECHOE 182.4, which survives only in a heavily damaged manuscript, but whose Latin source is known; a different kind of challenge would be ECHOE 284x.a, which survives only as a narrow strip with most of the text gone, but which is not a unique copy, so it is easier to reconstruct the text using other witnesses, as Page has done and as is reflected in the ECHOE edition.

Metrical Analysis

While there may be Latin XML corpora out there that encode metre, I am not aware of any such corpora of Old English or Old Norse, for instance, other than CLASP and the Electronic *Beowulf*, neither of which shares its metrical data in a downloadable format; but at least they are (somewhat) helpful for human reference. This state of affairs raises the question how much metrical information can be inferred from just the lines of text themselves. ASPR encodes the caesura between halflines, so we can analyze one halfline at a time, as demonstrated in the *n*-grams demo notebook. What information can we glean from there?

The first step to metrical analysis is syllabification. An early version of the CLTK library included a syllabifier for Old English, which you could try out; but at least if your aim is a straightforward syllable count, you would be better off writing your own function, as demonstrated in the corresponding demo notebook. Even so, a vowel-based syllable counter cannot account for complex rules of metrical prosody such as resolution (see my videos on Old English poetics for this concept and others). A function that identifies metrical lifts and Sievers types (again, see my videos) would be considerably more challenging yet; perhaps the easiest solution would be a corpus of individual words encoded with their possible metrical analysis, and a function that predicts a halfline's likeliest metrical type based on such an index. Either way, there is sufficient challenge to such an undertaking as to make it tantalizing. Of course if you choose to work with a corpus employing end-rhyme rather than alliteration you face a different set of challenges, such as the identification of homophonous sequences spelled differently.

Concordances and Synoptic Editions

The Helix project took an annotated corpus representing one manuscript copy of a text (the Old Low German *Heliand*), and was able to transfer much of its metadata to transcriptions of other manuscript witnesses of the same text. This was possible because the text in question was a poem with consistent line numbering and even division into halflines, so the code merely had to identify matching words within the halfline, i.e. within a group of just a few words at a time. It is these metadata that make the search engine on the Helix website especially useful, as the engine returns all forms of a word, not just identical forms. But the synoptic edition below it — i.e. the line-by-line comparison of the manuscript witnesses — requires no such metadata and is just a matter of displaying the lines in matching groups. And even a group of manuscript witnesses without any metadata may be analyzed for variation within the verse line as long as their order is consistent.

Perhaps you can think of another poem (by Chaucer, for instance) existing in multiple witnesses. Perhaps multiple copies are freely available online but without metadata, enabling you to write a script to identify textual variation within the line or perhaps even to identify tokens belonging to the same lemma across the text. Perhaps you could write a first-pass lemmatizer or part-of-speech tagger, even if the results will need to be corrected by hand. Or maybe you know of an annotated text whose metadata you could transfer onto a different manuscript copy of the same text. You wouldn't have to write a web-ready concordance or synoptic edition like Helix (which was written in PHP but accesses JSON files prepared in Python); it would suffice to demonstrate the principles in a Python notebook by way of Python functions. But of course I won't stop you if you want to go the extra mile.

Modern Corpora

If you would prefer to work with an early Modern (English or otherwise) corpus for greater accessibility, my suggestions may not be as detailed as for the above research angles, but I can offer a few rough ideas that may easily be adapted from textbooks (Bird et al.; Lane and Dyshel) or online tutorials:

- A cluster analysis of a corpus of your choice, using TF-IDF, document embeddings, or any other metric to infer which sets of documents within a corpus group together by that metric (accompanied by a discussion of what might explain your findings);
- Authorship analysis (stylometry): use any set of metrics (Burrows's delta; distribution of function words; etc.) deemed to define authorial profile, and then perform cluster analysis to see which documents group together, whether or not in order to uncover the author of some anonymous work;
- Shared formulas between the corpora of Shakespeare and Marlowe;
- A sentiment analysis classifier indicating of any early Modern English poem whether it is more cheerful or more downcast;
- A topic classifier capable of categorizing any early Modern English poem by its subject matter.